# Assignment- 14.5

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**BATCH – 04**

## TASK 1– Image Gallery Website

**Create an image gallery web page.**

**Requirements:**

- **Display images in a grid layout.**
- **Use CSS Grid.**
- **Add hover zoom effect.**
- **Make it mobile responsive.**

## PROMPT

**Generate responsive HTML and CSS code for an image gallery using CSS Grid with hover zoom effects and mobile responsiveness.**

## CODE

```
<!DOCTYPE html>
<html>
<head>
<title>Image Gallery</title>
<style>
body{
font-family: Arial;
text-align:center;
}
.gallery{  display:g
rid;
grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
gap:15px;
padding:20px;
}
.gallery
img{  width:100%
;
```

```
border-radius:10px;
transition:0.3s;
}
.gallery
img:hover{  transform:scale(1.1);
}
</style>
</head>
<body>
<h2>Image Gallery</h2>
<div class="gallery">
<img src="https://picsum.photos/300/200?1">
<img src="https://picsum.photos/300/200?2">
<img src="https://picsum.photos/300/200?3">
<img src="https://picsum.photos/300/200?4">
</div>
</body>
</html>
```

```html
<!DOCTYPE html>
<html>
<head>
<title>Image Gallery</title>

<style>

body{
font-family: Arial;
text-align:center;
}

.gallery{
display:grid;
grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
gap:15px;
padding:20px;
}

.gallery img{
width:100%;
border-radius:10px;
transition:0.3s;
}

.gallery img:hover{
transform:scale(1.1);
}

</style>

</head>

<body>

<h2>Image Gallery</h2>

<div class="gallery">

<img src="https://picsum.photos/300/200?1">
<img src="https://picsum.photos/300/200?2">
<img src="https://picsum.photos/300/200?3">
<img src="https://picsum.photos/300/200?4">

</div>

</body>
</html>
```

## RESULT

**Images are displayed in a grid format.**

## OUTPUT

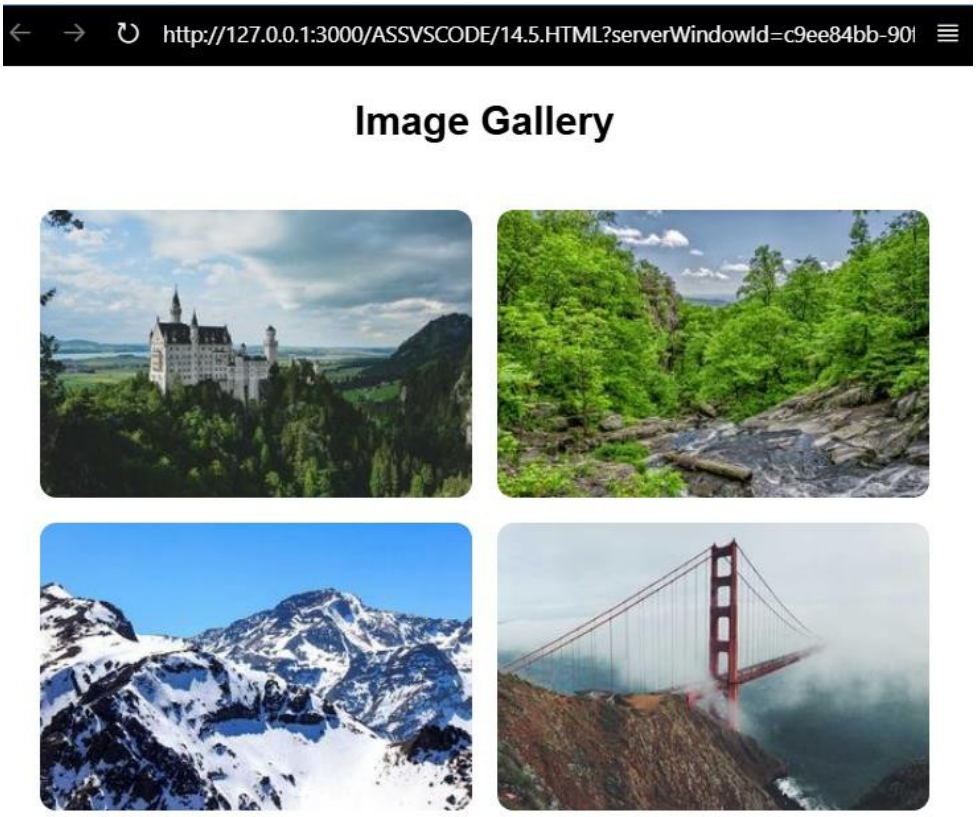



**Image gallery appears with hover zoom effect.**

## EXPLANATION

**CSS Grid arranges the images automatically in rows and columns. The hover transform enlarges images slightly to create an interactive UI.**

## TASK 2 – Profile Card Generator

## QUESTION

**Create a profile card displaying:**

- **User image**
- **Name**
- **Role**
- **Social links**

**Using Flexbox with hover animation.**

## PROMPT

**Generate a modern profile card using HTML and CSS Flexbox with hover effects and shadows.**

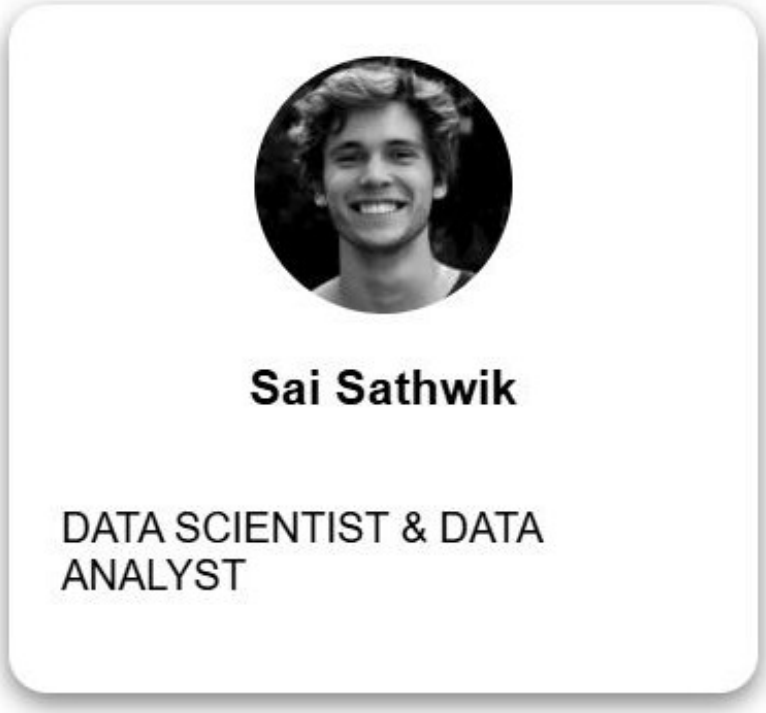
**CODE**

```html
<!DOCTYPE html>
<html>
<head>
<style>
body{
font-family:Arial;
display:flex;
justify-content:center;
margin-top:50px;
}
.card{
width:250px;
padding:20px;
border-radius:15px;
box-shadow:0 4px 10px gray;
display:flex;
flex-direction:column;
align-items:center;
transition:0.3s;
}
.card:hover{
transform:translateY(-8px);
}
img{
width:100px;
border-radius:50%;
}
</style>
</head>
<body>
<div class="card">
<img src="https://i.pravatar.cc/150">
<h3>Sai Sathwik</h3>
<p>DATA SCIENTIST & DATA ANALYST</p>
</div>
</body>
</html>
```

**RESULT**

**A modern profile card appears.**

## OUTPUT




**Card lifts upward when hovering.**

## EXPLANATION

**Flexbox aligns elements vertically while hover animation improves visual interaction.**

## TASK 3 – Dynamic Shopping Cart System

## QUESTION

**Develop a shopping cart system that:**

- **Adds products**
- **Removes products**
- **Updates quantity**
- **Displays real-time total price**
- **Stores data in local Storage.**

## PROMPT

**Generate JavaScript code for a shopping cart with add/remove functionality and total price calculation using local Storage.**

## CODE

```html
<!DOCTYPE html>
<html>
<head>
<title>SHOPPING BASKET</title>
</head>
<body>
<h2>Shopping Cart</h2>
<button onclick="addItem('Product',100)">Add Product</button>
<p>Total Price: ₹<span id="total">0</span></p>
<script>
let cart=[];
function addItem(name,price){
cart.push({name,price});
updateTotal();
}
function updateTotal(){
let total=cart.reduce((sum,item)=>sum+item.price,0);
document.getElementById("total").innerText=total;
localStorage.setItem("cart",JSON.stringify(cart));
}
</script>
</body>
</html>
```

## RESULT

**Products can be added and total price updates automatically.**

## OUTPUT

# SHOPPING BASKET

Add Product

Total Price: ₹0

**Total price increases when items are added.**

## EXPLANATION

**JavaScript objects store product data and local Storage preserves cart information.**

## TASK 4 – Role Based Access Control

## QUESTION

**Create roles:**

- **Admin**
- **Editor**
- **User**

**Different UI should appear for each role.**

## PROMPT

**Generate JavaScript logic for role-based access control with conditional UI rendering.**

## CODE

```
<!DOCTYPE html>
<html>
<body>
<h2>Role Based Dashboard</h2>
<button onclick="showDashboard('admin')">Admin</button>
<button onclick="showDashboard('editor')">Editor</button>
<button onclick="showDashboard('user')">User</button>
<p id="output"></p>
<script>
function showDashboard(role){
if(role=="admin"){
document.getElementById("output").innerText="Admin Dashboard";
}
else if(role=="editor"){
document.getElementById("output").innerText="Editor Dashboard";
}
else{
document.getElementById("output").innerText="User Dashboard";
}
}
</script>
</body>
</html>
```

## RESULT

**Different dashboards appear depending on role.**

## OUTPUT

# Role Based Dashboard

Admin Editor User

User Dashboard

# Role Based Dashboard

Admin Editor User

Editor Dashboard

# Role Based Dashboard

Admin Editor User

Admin Dashboard

**Admin / Editor / User dashboard message displays.**

## Explanation

**Conditional statements check the role and render the appropriate interface.**

## TASK 5 – Multi Language Website

## QUESTION

**Create a website that:**

- **Switches languages**
- **Stores language preference.**

## PROMPT

**Generate JavaScript code for dynamic language switching with local Storage.**

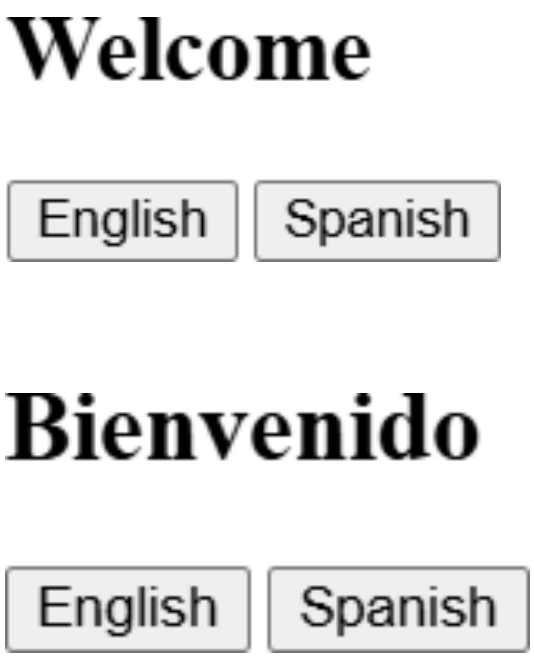
## CODE

```html
<!DOCTYPE html>
<html>
<body>
<h2 id="title">Welcome</h2>
<button onclick="setLanguage('en')">English</button>
<button onclick="setLanguage('es')">Spanish</button>
<script>
const translations={
en:{title:"Welcome"},
es:{title:"Bienvenido"}
}
function setLanguage(lang){
document.getElementById("title").innerText=translations[lang].title;
localStorage.setItem("lang",lang)
}
</script>
</body>
</html>
```

**RESULT**

**Language changes dynamically.**

**OUTPUT**




**Title changes between English and Spanish.**

**EXPLANATION**

**Translation objects store text for each language and JavaScript updates the UI.**

## Task 6 – Online Examination System

## QUESTION

**Create an exam interface with:**

- **Timer**
- **Questions**
- **Submission**
- **Prevent multiple submissions.**

## PROMPT

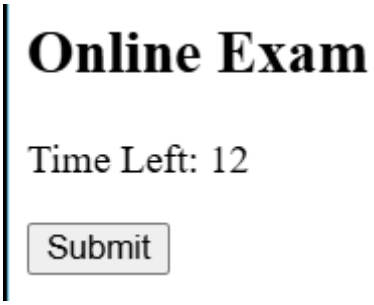
**Generate JavaScript code for an exam timer with automatic submission.**

```html
<!DOCTYPE html>
<html>

<body>

<h2>Online Exam</h2>

<p>Time Left: <span id="timer">30</span></p>

<button onclick="submitExam()">Submit</button>

<script>

let time=30;

let timer=setInterval(function(){

document.getElementById("timer").innerText=time;

time--;
if(time<0){
clearInterval(timer);
submitExam();
}
},1000);
function submitExam(){
alert("Exam Submitted");
}
</script>
</body>
</html>
```

**RESULT**

**Timer counts down and exam is submitted automatically.**

**OUTPUT**




**Alert message shows "Exam Submitted".**

**EXPLANATION**

**A countdown timer controls exam duration and ensures automatic submission.**